

# Roof Lifecycle Cost Analysis

Laurel market  
October 6, 2026

Planned maintenance is projected to save the owner of a typical Laurel school **\$1.29M** over the next 20 years.

About 35% less than reactive repair: \$940K lower spending plus \$352K more roof life at year 20. Range \$643K–\$1.70M from conservative to upside assumptions; \$674K in today's dollars at a 7% discount rate.

| MARKET | PROPERTY | ROOF AREA     | SYSTEM           | AGE · CONDITION | NEW ROOF TODAY |
|--------|----------|---------------|------------------|-----------------|----------------|
| Laurel | K-12     | 120,000 sq ft | Modified bitumen | 8 yrs · Good    | \$960K         |

**PAXSEAL NET COST**

**\$2.39M**

\$2.29M spent + \$576K roof used – \$480K roof left

**REACTIVE NET COST**

**\$3.68M**

\$3.23M spent + \$576K roof used – \$128K roof left

**REPAIR & STORM SAVINGS**

**\$1.58M**

Unbudgeted repairs and storm damage avoided

**ROOF PRICES**

**2× in 15 yrs**

At 4.6%/yr, the long-run trend in commercial roofing costs

Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

|                           | PAXSEAL | REACTIVE | SAVINGS  |
|---------------------------|---------|----------|----------|
| PaxSeal fees              | \$441K  | —        | -\$441K  |
| Repairs paid by owner     | \$234K  | \$1.79M  | +\$1.56M |
| Expected storm damage     | \$40K   | \$60K    | +\$20K   |
| Roof replacements         | \$1.57M | \$1.38M  | -\$190K  |
| Cash spending             | \$2.29M | \$3.23M  | +\$940K  |
| Current roof value used   | \$576K  | \$576K   | —        |
| Roof life left at year 20 | -\$480K | -\$128K  | +\$352K  |
| Net 20-year cost          | \$2.39M | \$3.68M  | +\$1.29M |

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
- 2 **Smaller repairs.** Emergency repairs run about 3.9× the planned rate in a school, and the PaxSeal fee covers half of planned repairs. Together that saves \$1.56M.
- 3 **A predictable budget line.** Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$3.13 of other cash roof costs.

Local risk profile

Prince George's County, MD (24033)

Thunderstorm wind / derecho, snow load

About 3–4 major regional roof-damaging events per 20 years, such as the 2010 snowstorms, the 2012 derecho and the 2016 blizzard. Expected storm damage to this roof over 20 years is **\$40K** if maintained and **\$60K** with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

|                                 |                     |       |
|---------------------------------|---------------------|-------|
| Hail                            | Relatively Moderate | ■■■■■ |
| Strong wind                     | Relatively High     | ■■■■■ |
| Tornado                         | Relatively High     | ■■■■■ |
| Hurricane                       | Relatively Moderate | ■■■■■ |
| Winter weather                  | Very High           | ■■■■■ |
| Ice storm                       | Relatively Moderate | ■■■■■ |
| FEMA National Risk Index rating |                     |       |

Key assumptions

The basis for each figure in this analysis.

| ASSUMPTION           | VALUE            | BASIS                                                                                         |
|----------------------|------------------|-----------------------------------------------------------------------------------------------|
| Roof life            | 20 / 15 yrs      | Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties) |
| Roof cost growth     | 4.6%/yr          | BLS Producer Price Index, nonresidential roofing, 2007–2025                                   |
| Replacement cost     | \$8.00/sq ft     | Laurel contractor price ranges                                                                |
| Reactive repair cost | 3× planned × 1.3 | Industry estimates; ×1.3 for leak damage (school). PaxSeal fee covers half of planned repairs |
| Storm damage         | 0.13%/yr         | Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index                    |
| PaxSeal fee          | \$16,400/yr      | Year 1, rising 3%/yr. Estimate; your proposal states the actual fee                           |

**Next step:** a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email [info@paxserv.com](mailto:info@paxserv.com).

**Estimates, not a quote.** Projections from Laurel market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance.